Illustration

CPR Coach

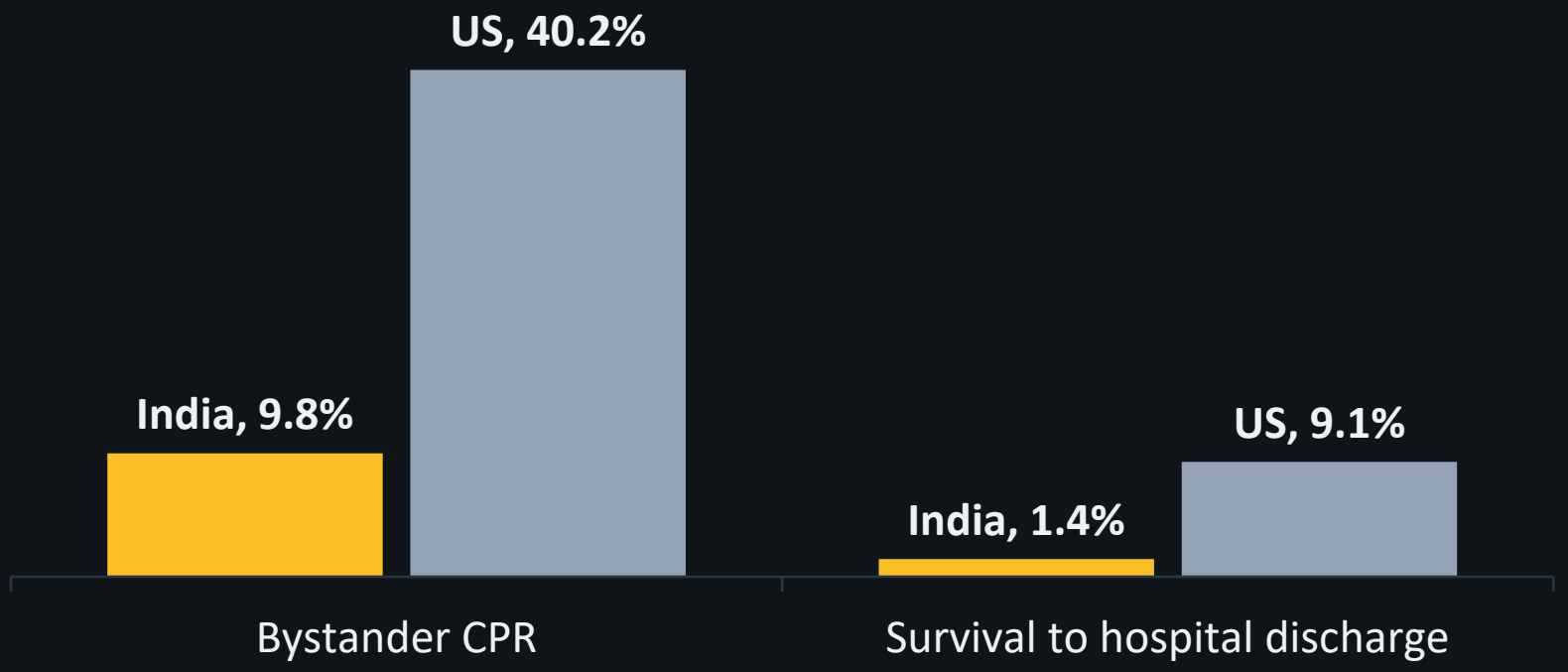
A phone that watches your CPR and tells you what to fix.



Team Zero Flux Vivek · Pratham · Prabal · Moksh

ANVATHON 2026 · OPEN INNOVATION

India's bystander CPR rate is a quarter of the US rate



India
one hospital, 205 patients

US
national registry

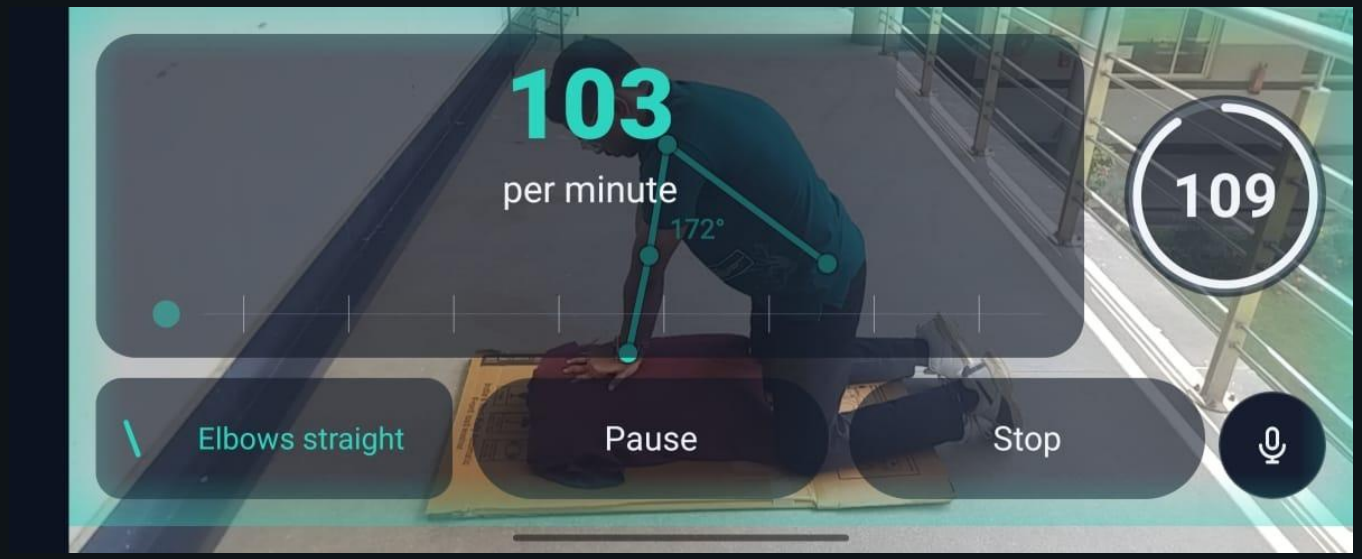
Fewer people nearby start CPR, and fewer patients leave hospital alive.

Karnataka is moving: in July 2025 the Health Minister announced CPR training for school and college students, teachers and gym trainers.

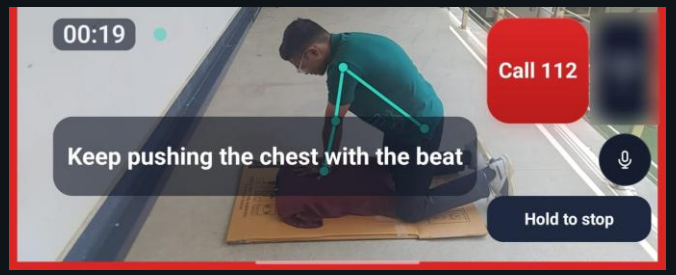
India: Bhat et al., Indian Heart Journal 2021, one tertiary hospital, 205 patients.
Karnataka: NewKerala, 10 July 2025.
The two sources are not measured the same way: one hospital vs a national registry.

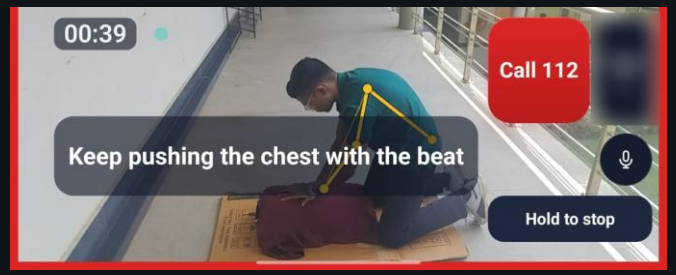
US: American Heart Association, Heart Disease and Stroke Statistics 2023.
Bystander CPR: over 356,000 out-of-hospital arrests. Survival: 2021 US data, EMS-treated adult non-traumatic arrests.

It watches your arms and says what to fix



Practise: 103 pushes a minute, elbows straight





Emergency: joints turn amber when posture needs fixing

Watch the camera finds shoulder, elbow and wrist

Count push rate, elbow angle and pauses

Speak one correction at a time, out loud



All you need: a phone and a home-made manikin.

Measures rate, elbows and shoulders over hands. Can't measure depth, force or hand placement.

Coaches in 10 languages, fully offline.

Camera to voice, every step runs on the phone



phone held side-on

MediaPipe finds the joints

rate, elbow angle, pauses

one correction at a time

recorded clips and a 110/min beat

No internet permission. Nothing is uploaded.

`Kotlin · Jetpack Compose · CameraX · MediaPipe Pose · sherpa-onnx (Whisper, IndicConformer) · Silero VAD`

Illustration

Built in 24 hours: 596 commits

FROM

Our first real run on a phone crashed mid-session, and the recording was lost.

TO

A working prototype: the crash fixed, camera load capped, and heat and battery logged on every run.

399

tests passing, 0 failing

7

real phone recordings must replay byte for byte

Illustration

Live demo

Next: hand-counted accuracy runs, review by CPR trainers, college pilots.

Not a medical device. In an emergency, call 112 (108 for an ambulance) and start CPR.

Code and test records

`github.com/VivekTiwary-25/Zero_flux`

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